ServiceNow Development

Project Documentation format

1. Introduction

•**Project Title:** Import Data using Transform Maps

•**Team Members:**

Team Member	Role	Responsibility
Naresh S	Data & Table Developer	Prepared the sample spreadsheet and created the Employee Test table
Vignesh S	Import & Transform Developer	Created the Import Set table and Transform Map; ran and validated the transform
Praveen Kumar G	Data Integrity Developer	Enabled Coalesce on Employee ID and tested repeat imports with new Excel data
Praveen Kumar G	Reports & Dashboard Developer	Created the reports and the Employee Analytics Dashboards

2. Project Overview

•**Purpose:** This micro project imports structured employee data from an external Excel spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. It simulates a real-world case where bulk employee data arrives as a spreadsheet and has to be migrated accurately. The data is first loaded into an Import Set staging table, a Transform Map moves it into the target table, and the result is then verified. The project helps beginners understand the end-to-end import process: data loading, field mapping, transformation and result verification.

- **Features:**

•Spreadsheet (.xlsx) data loaded through Load Data into the Employee Import (u_employee_import) staging table.

•Custom target table, Employee Test (u_employee_test), with Employee ID, Employee Name, Email, Department and Location fields.

•Transform Map with automatic and manual field mapping between the staging and target tables.

•Coalesce on Employee ID so that repeated imports update existing records instead of inserting duplicates.

•Transform History showing total, inserted, updated, ignored and error counts for every run.

•Reports (Employees by Department, Employees by Location, Employee List Report) and an Employee Analytics Dashboards page.

3. Architecture

The project runs entirely on the ServiceNow platform. Data moves through the stages below.

Excel Spreadsheet	→	Load Data	→	Import Set Table	→	Transform Map	→	Employee Test Table

- **Frontend:** The ServiceNow web interface is used instead of a custom React frontend. Forms and lists (Table, Load Data, Transform Map, Employee Tests), the Report Designer (Create a report) and the Dashboards page (PA_DASHBOARDS) provide the user interface.
- **Backend:** The server-side logic is provided by ServiceNow Import Sets and Transform Maps. Load Data copies each spreadsheet row into the staging table, the Transform Map applies the field maps (with Coalesce on Employee ID) and writes the rows to the target table. Access Control Rules (ACLs) protect the tables and the dashboard form.
- **Database:** ServiceNow stores data in relational tables. The two custom tables are described below.

Table	Field (Column Label)	Type	Purpose
Employee Import (u_employee_import)	Employee ID, Name, Email, Department, Location	String	Staging table that holds the raw spreadsheet rows
Employee Test (u_employee_test)	Employee ID, Employee Name, Email, Department, Location	String (max length 40)	Target table that stores the final employee records

4. Setup Instructions

- **Prerequisites:**

- A ServiceNow instance (for example a Personal Developer Instance) and a user with the admin role.
- Google Sheets or Microsoft Excel to prepare the .xlsx file.
- A modern web browser.

- **Installation:** No software has to be installed. Set up the project as follows.

1. Log in to the ServiceNow instance as an administrator.
2. Create a Google Spreadsheet with the columns Employee ID, Name, Email, Department and Location, add sample data, and download it as Sample Spreadsheet.xlsx.
3. Create the custom table Employee Test (u_employee_test) with the five String fields (Tables > Create New).
4. Open Load Data, create the import table Employee Import (u_employee_import) from the .xlsx file, then click Create transform map.
5. Create the Transform Map Sample Spreadsheet Import from Employee Import to Employee Test and use Auto Map Matching Fields.

5. Folder Structure

ServiceNow has no client/server folders. The project components are organised as the following records on the platform.

- **Client (user-facing components):**

Component	Type	Whereto findit
Employee Tests Employees by Department (and Location, List Report) Employee Analytics Dashboards	List / Form	All menus > Employee Test
	Reports Dashboard	All menus > Reports (Usage and governance)
		All menus > Dashboards (Self Service)

- **Server (back-end components):**

Component	Type	Where to find it
Employee Import (u_employee_import)	Import Set (staging) table	System Import Sets > Load Data
Sample Spreadsheet Import	TransformMap (withCoalesce)	System Import Sets > Transform Maps
Employee Test (u_employee_test)	Custom target table	Tables
pa_dashboards (create)	Access Control Rule	System Security > Access Control (ACL)

6. Running the Application

Run the import from the ServiceNow instance:

- **Load the data:** All menus > Load Data. Choose Existing table (Employee Import), select the .xlsx file, set Sheet number 1 and Header row 1, and click Submit.
- **Run the transform:** Under Next steps click Run Transform, select the Transform Map and click Transform. The state should show Complete with completion code Success.
- **Verify:** Open Employee Test to view the records, and open Transform History to check the inserted, updated and ignored counts.
- **View the results:** Open Dashboards and select Employee Analytics Dashboards.

7. API Documentation

No custom API was built. The standard ServiceNow REST APIs are available for the tables created in this project and are listed here for reference; the project itself used the Load Data and Transform screens.

Method	Endpoint	Description
GET	/api/now/table/u_employee_test	Returns records from the Employee Test table (Table API)
POST	/api/now/table/u_employee_test	Creates one Employee Test record (Table API)
POST	/api/now/import/u_employee_import	Inserts a row into the staging table (Import Set API)

Example response for GET /api/now/table/u_employee_test?sysparm_limit=1:

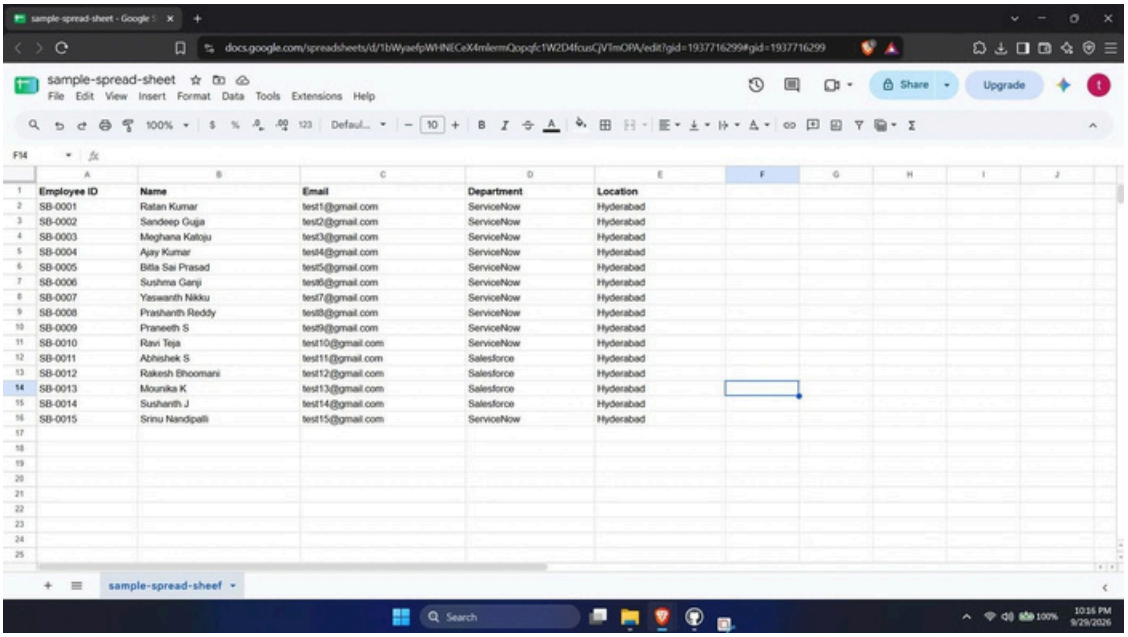
{ "result": [{ "u_employee_id":"SB-0001", "u_employee_name":"Ratan Kumar", "u_email": "test1@gmail.com", "u_department": "ServiceNow", "u_location": "Hyderabad" }] }

8. Authentication

- Users sign in to the ServiceNow instance with their credentials; the platform manages sessions and tokens. REST calls use basic authentication or OAuth.
- Authorization uses roles and Access Control Rules (ACLs). The project was configured and tested with an administrator account.
- To open the dashboard form directly with the backend name PA_DASHBOARDS.FORM, the create ACL for pa_dashboards needs Admin overrides set to true.
- The Employee Analytics Dashboards record is owned by System Administrator and can be restricted by users, groups and roles from the Share option.

9. User Interface

Milestone 1 – Spreadsheet and table creation



Employee ID	Name	Email	Department	Location
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Guja	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad

Figure 1: Sample spreadsheet with employee data

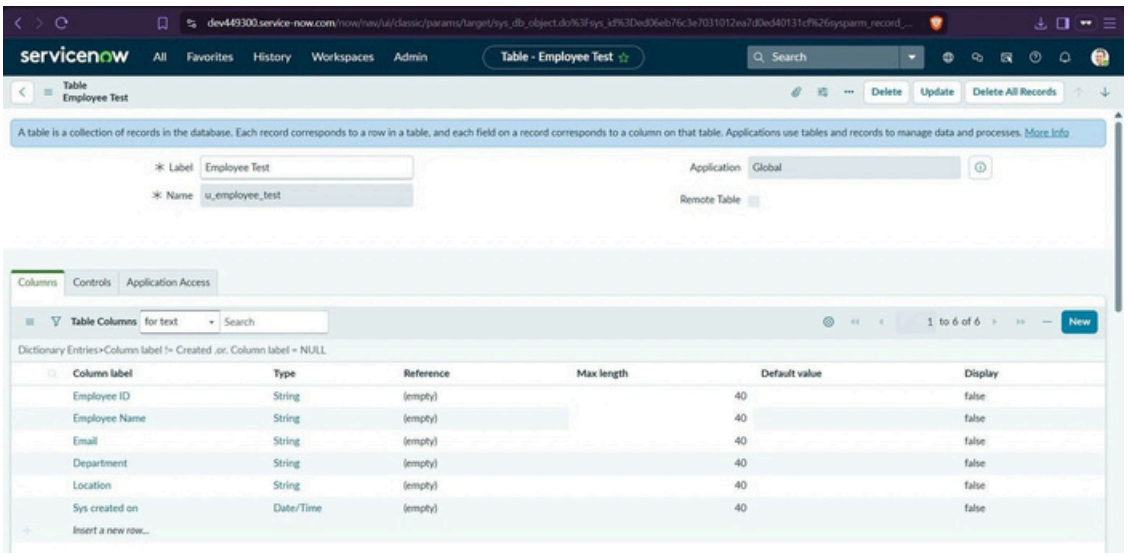


Figure 2: Employee Test table and its columns

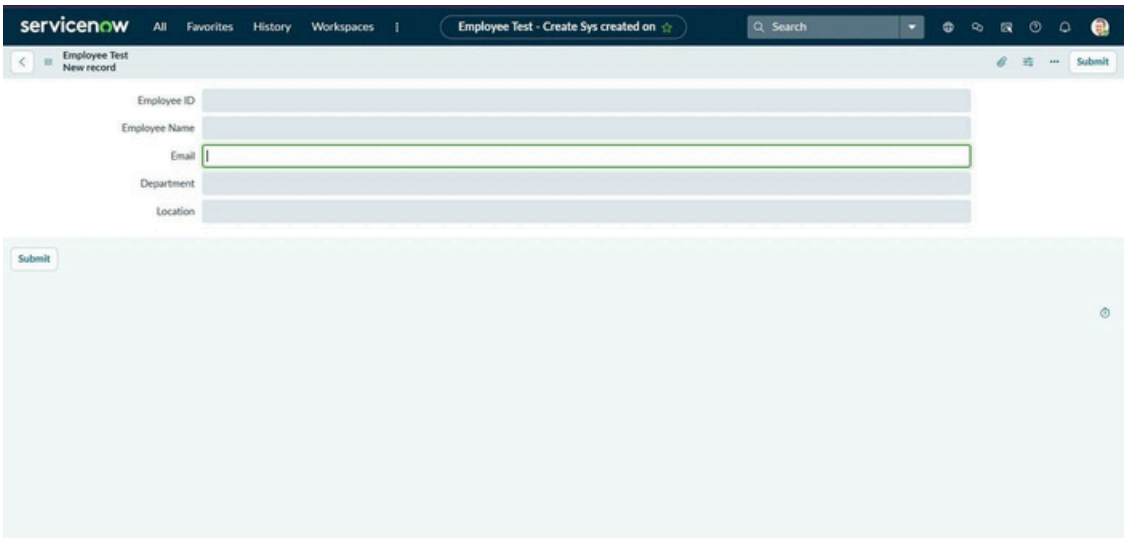


Figure 3: Employee Test form showing the new fields

Milestone 2 – Import Set table and Transform Map

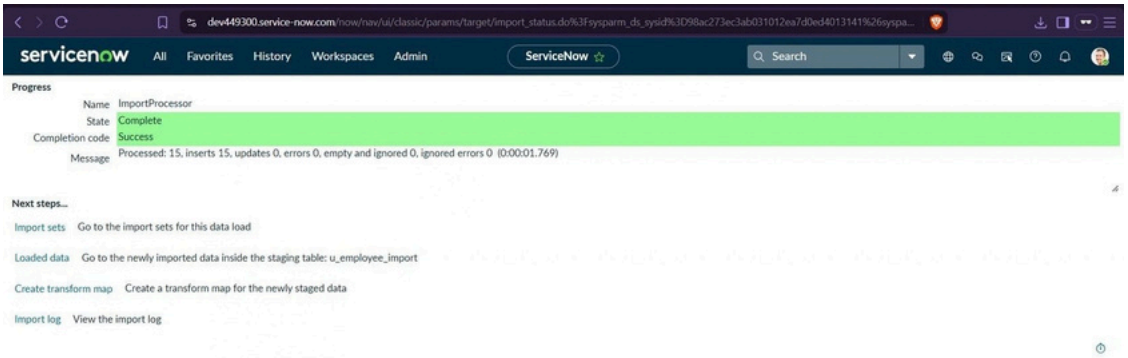


Figure 4: Load Data completed successfully (15 rows inserted into the staging table)



Figure 5: Mapping Assist showing source and target field mapping

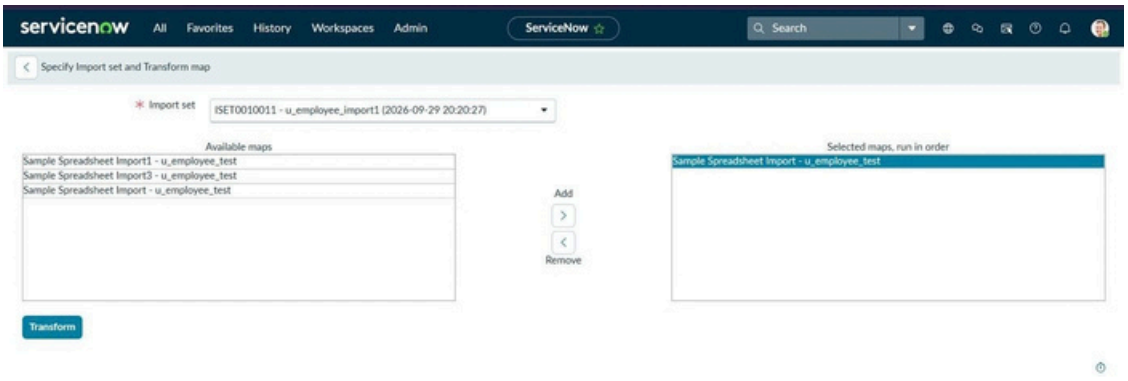
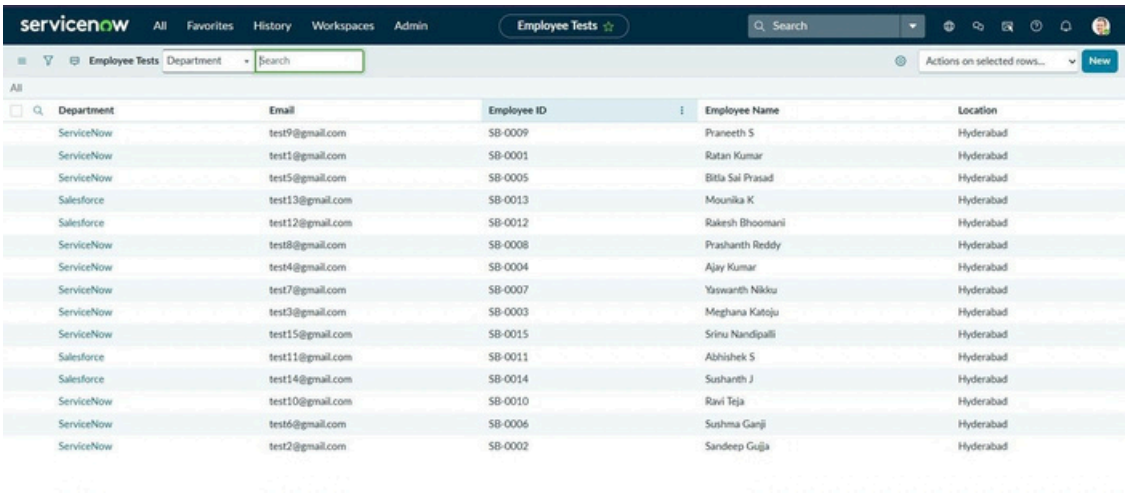


Figure 6: Selecting the import set and Transform Map to run



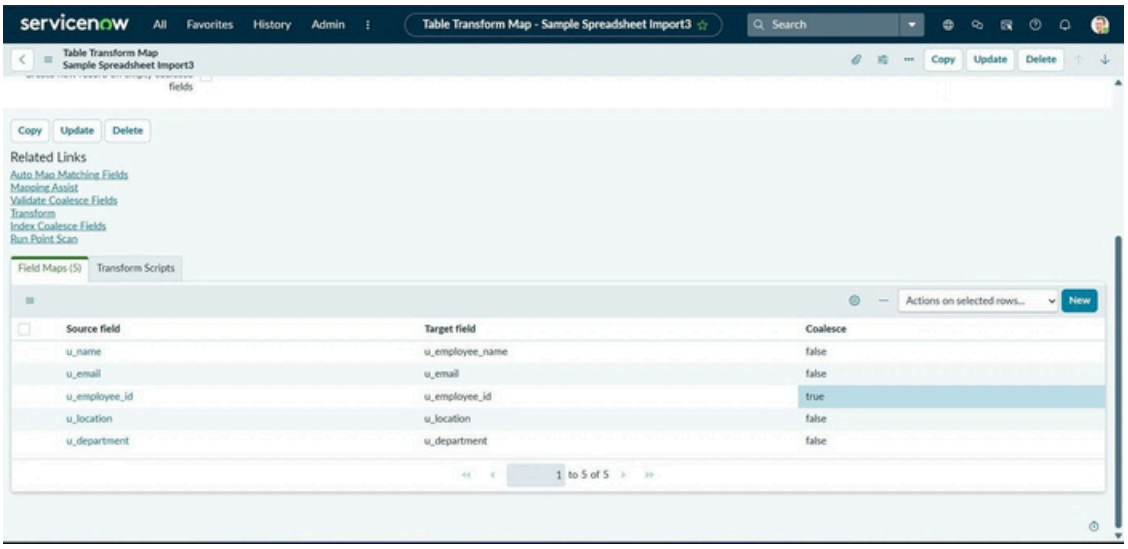
Figure 7: Transformation completed with state Complete and code Success

Milestone 3 – Transform, validate and Coalesce



Department	Email	Employee ID	Employee Name	Location
ServiceNow	test9@gmail.com	SB-0009	Praneeth S	Hyderabad
ServiceNow	test1@gmail.com	SB-0001	Ratan Kumar	Hyderabad
ServiceNow	test5@gmail.com	SB-0005	Bitla Sai Prasad	Hyderabad
Salesforce	test13@gmail.com	SB-0013	Mounika K	Hyderabad
Salesforce	test12@gmail.com	SB-0012	Rakesh Bhoomani	Hyderabad
ServiceNow	test8@gmail.com	SB-0008	Prashanth Reddy	Hyderabad
ServiceNow	test4@gmail.com	SB-0004	Ajay Kumar	Hyderabad
ServiceNow	test7@gmail.com	SB-0007	Yaswanth Nikku	Hyderabad
ServiceNow	test3@gmail.com	SB-0003	Meghana Katoju	Hyderabad
ServiceNow	test15@gmail.com	SB-0015	Srinu Nandipalli	Hyderabad
Salesforce	test11@gmail.com	SB-0011	Abhishek S	Hyderabad
Salesforce	test14@gmail.com	SB-0014	Sushanth J	Hyderabad
ServiceNow	test10@gmail.com	SB-0010	Ravi Teja	Hyderabad
ServiceNow	test6@gmail.com	SB-0006	Sushma Ganji	Hyderabad
ServiceNow	test2@gmail.com	SB-0002	Sandeep Gujja	Hyderabad

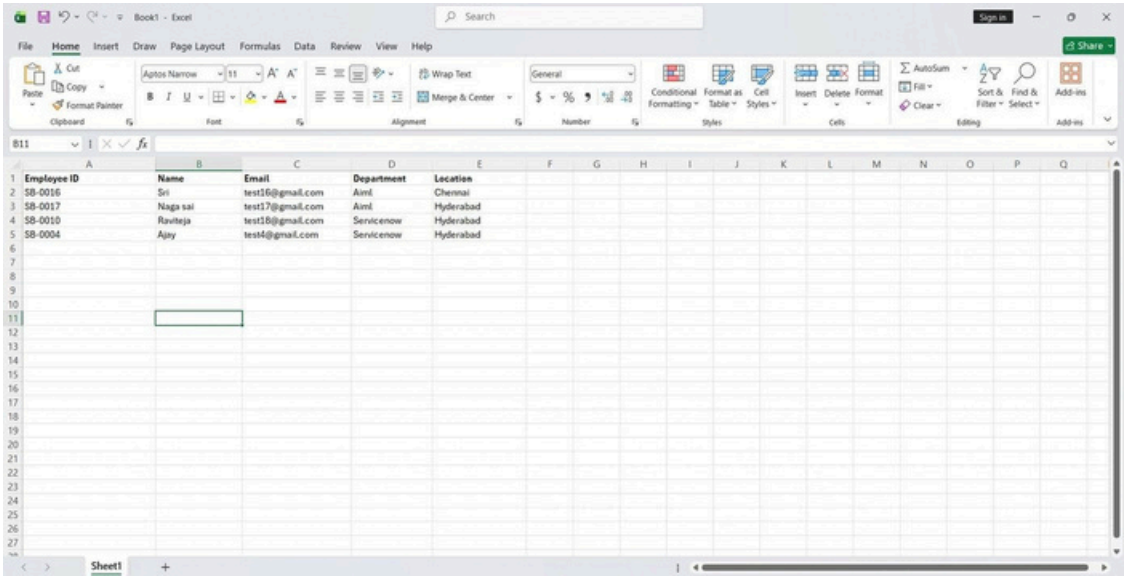
Figure 8: Employee Test records after the first import (15 records)



Source field	Target field	Coalesce
u_name	u_employee_name	false
u_email	u_email	false
u_employee_id	u_employee_id	true
u_location	u_location	false
u_department	u_department	false

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Figure 9: Coalesce set to true on the Employee ID field map



Employee ID	Name	Email	Department	Location
SB-0016	Sri	test16@gmail.com	Aiml	Chennai
SB-0017	Naga sai	test17@gmail.com	Aiml	Hyderabad
SB-0010	Raviteja	test18@gmail.com	Servicenow	Hyderabad
SB-0004	Ajay	test4@gmail.com	Servicenow	Hyderabad

Figure 10: New Excel data used for the second import

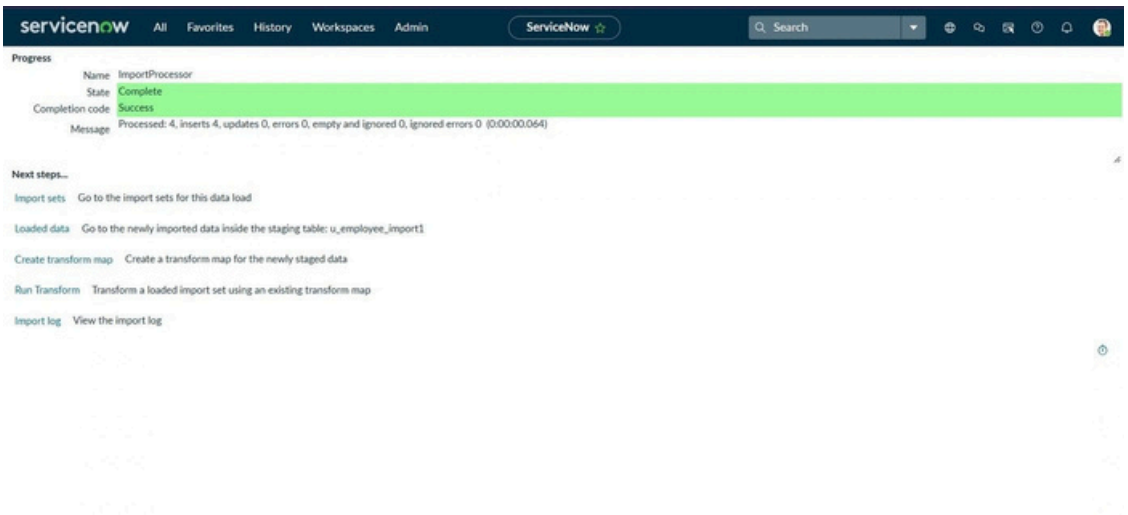


Figure 11: Load Data result for the new Excel file (4 rows)

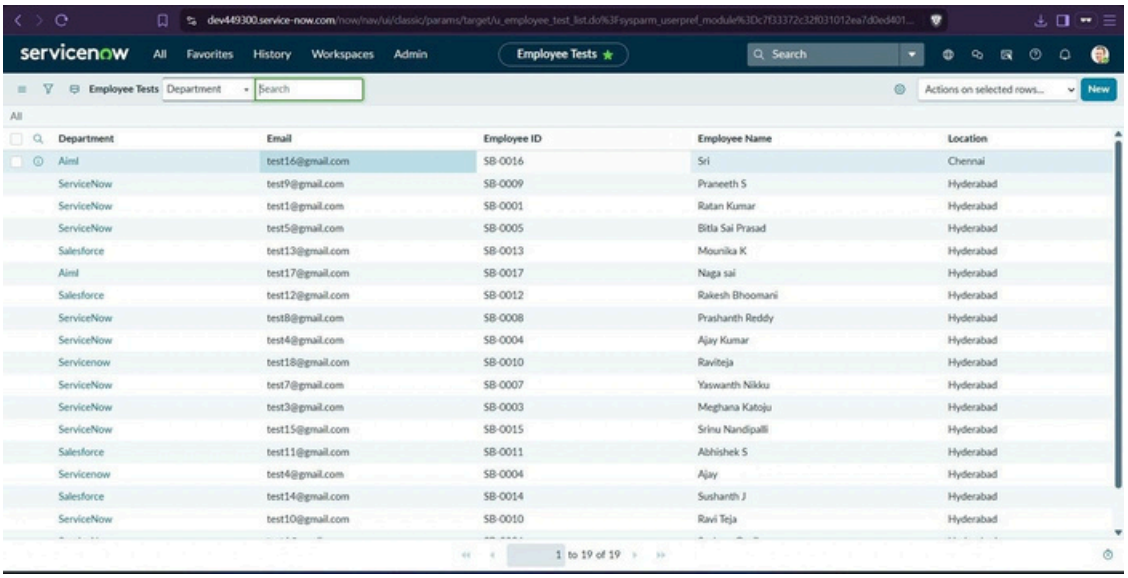


Figure 12: Employee Test records after the second import

10. Testing

- **Strategy:** The project was tested manually at each milestone, followed by performance testing of the import and User Acceptance Testing (UAT).
- **Tools:** ServiceNow Transform History, Import Log, list views of the Employee Test table, and the Reports and Dashboards pages.
- **Checks:** Successful load (state Complete / Success), field mapping, correct record counts (inserted, updated, ignored, errors), duplicate prevention with Coalesce, and report and dashboard output.
- **Test documents:** See the Performance Testing and User Acceptance Testing documents for the detailed test cases.

11. Screenshots or Demo

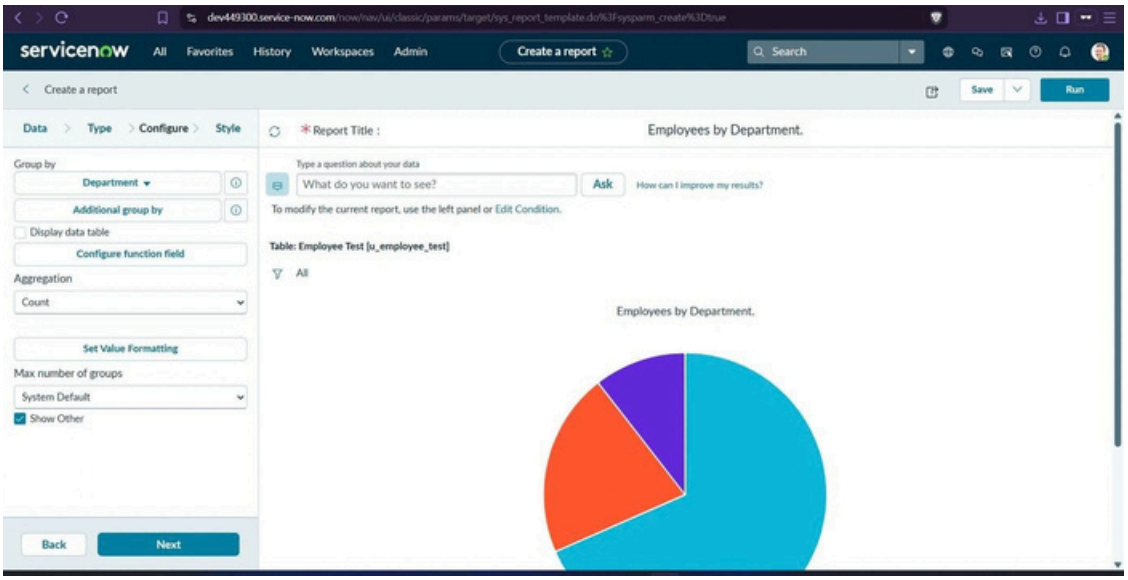


Figure 13: Report configuration: group by Department with Count aggregation

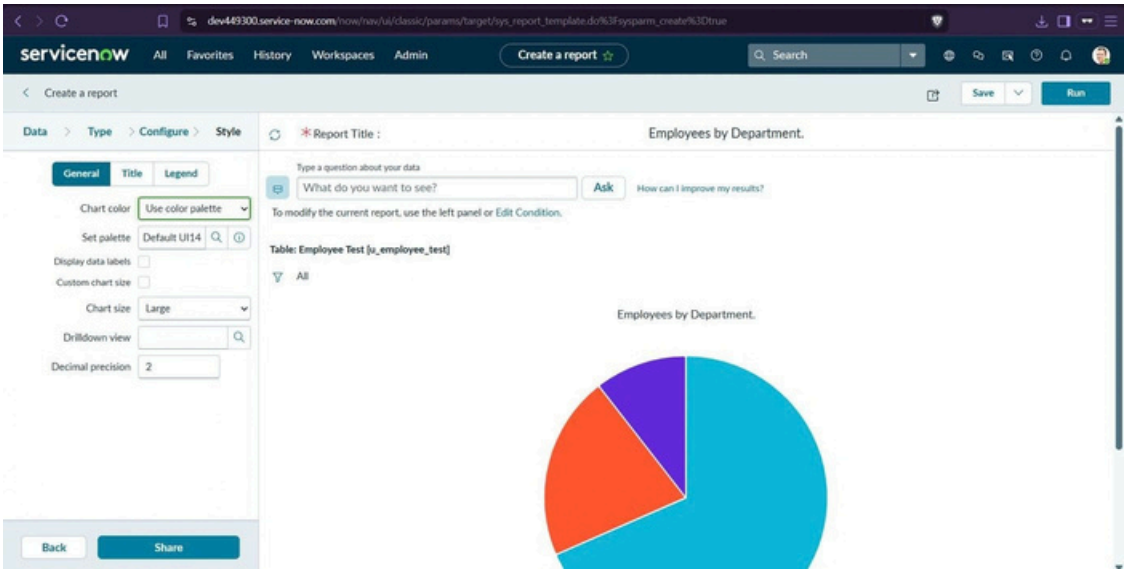


Figure 14: Employees by Department pie chart (style settings)

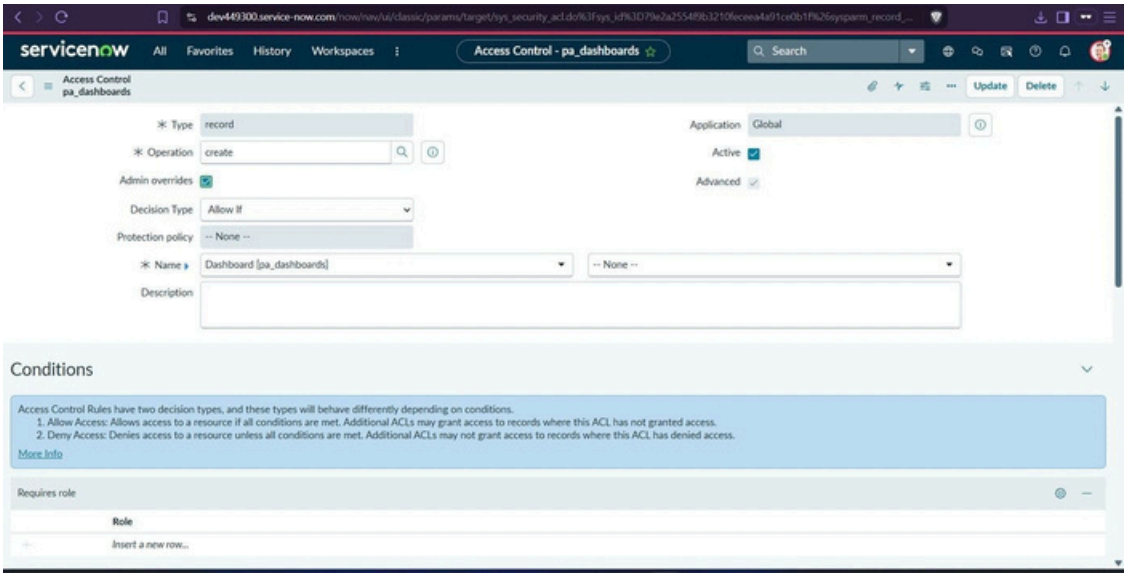


Figure 15: Create ACL for pa_dashboards with Admin overrides enabled

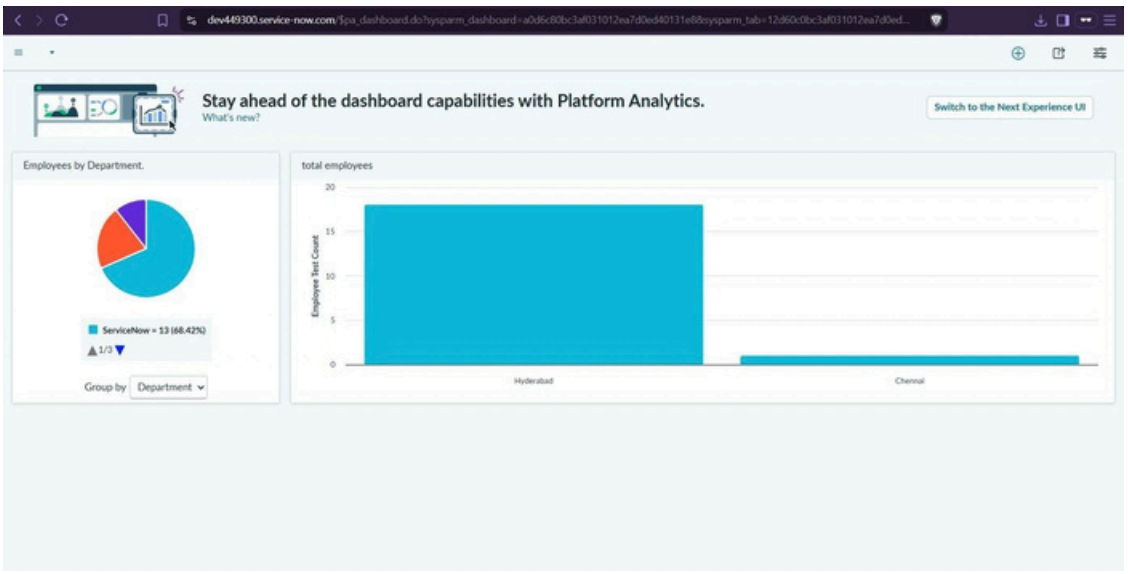


Figure 16: Employee Analytics Dashboards showing the department and location reports

Demo link: https://dev449300.servicenow.com/$pa_dashboard.do?sysparm_dashboard=a0d6c80bc3af031012ea7d0ed40131e

12. Known Issues

- Duplicate records after the second import: the Employee Test list shows 19 records (15 initial + 4 from the modified file), so Employee IDs SB-0004 and SB-0010 appear twice instead of being updated. This means Coalesce on Employee ID was not applied in that transform run; the second import used a Transform Map that was selected from several similar maps (Sample Spreadsheet Import, Import1, Import3). Delete the duplicates and re-run the transform with the map that has Coalesce enabled.
- Because the reports read from Employee Test, the Employees by Department chart and the dashboard totals currently include those duplicates.
- Department values are not consistent in case (for example "Servicenow" and "ServiceNow"), so the same department can appear as separate groups in a report.
- Only a small sample data set was tested; large imports were not measured.
- The dashboard form can only be opened by its backend name after the pa_dashboards create ACL has Admin overrides enabled.

13. Future Enhancements

- Add a transform script to validate data (trim spaces, standardise case, check the email format) before records are inserted.
- Convert Department and Location into reference fields with their own tables.
- Schedule imports automatically from a Data Source (REST, JDBC or an attached file) instead of manual Load Data.
- Use email as a second coalesce field and send notifications when an import has errors.
- Add role-based access to the reports and dashboard, and build a dashboard of import success and error counts.
- Support CSV files and larger data volumes with performance monitoring.